



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH	IIT Kharagpur	8.46 / 10
2024	B.Tech	COAE,UAS Bangalore	7.568 / 10
2019	CLASS XII,CBSE	TVN,Varanasi	68.6%
2017	CLASS X,CBSE	TVN,Varanasi	7.8 / 10

PROJECTS

- Real-Time Network Fault Detection on Streaming Telemetry** | Self Project
Python | Scikit-learn | TensorFlow/Keras | Pandas | Streamlit
- Built a real-time **anomaly detection pipeline** using streaming server telemetry and rolling-window features
 - Developed an **Isolation Forest + Autoencoder ensemble** for network fault detection under severe class imbalance
 - Achieved **100% fault-window detection** while maintaining approximately **2 false alarms per day** through operating-point tuning
 - Deployed a **streamlit NOC monitoring dashboard** with live anomaly alerts and telemetry visualization
- Customer Churn Prediction and Retention Analytics** | Self Project
Tabular Machine Learning | Python, Pandas, Scikit-learn, XGBoost
- Built an end-to-end **customer churn prediction pipeline** with data preprocessing, EDA, feature engineering, and ML modeling
 - Compared **Logistic Regression, Random Forest, and XGBoost** using F1-score, precision, Recall, and ROC-AUC
 - Identified key **churn drivers** and generated customer-level predictions to support targeted retention strategies
- Experimental Characterization and Mathematical Modeling of Carangiform BCF Locomotion in Nile Tilapia** | M.Tech Thesis (Aug'26-Present)
- Developing a **computer-vision-based workflow** to extract and reconstruct the **body midline and caudal-fin motion** of Nile tilapia from high-speed swimming video
 - Quantifying **BCF swimming kinematics**, including tail-beat frequency, lateral amplitude, wavelength, phase propagation, wave speed, and body curvature

INTERNSHIPS

- i4-Marine Technologies Pvt. Ltd., Pune | Marine Technology Intern** (May'26 -Jul'26)
- Designed and developed an **IoT-based Automatic Fish Feeder System** for automated feeding in aquaculture applications
 - Designed **hopper and mechanical components using CAD**, integrating **ESP32/Raspberry Pi, Pixhawk, sensors, and motors** to develop an automated feeding system
 - Performed hardware testing, system validation, and troubleshooting** to ensure reliable and consistent operation of the automated feeding system
 - Developed practical understanding of **ROVs, autonomous marine systems**, and marine technologies through exposure to design, operation, and field applications

SKILLS AND EXPERTISE

- Programming:** Python, SQL, MATLAB **ML/Data Science:** Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn
- Software &Tools:** AnyLogic, Power BI, MS Excel, MS SQL, SolidWork, ANSYS, QGIS **IoT &Embedded System:** ESP32, Raspberry Pi

POSITIONS OF RESPONSIBILITY

- General Secretary -Technology | Atal Bihari Vajpayee Hall of Residence,IIT-Kharagpur** (2025-2026)
- Led technology initiatives and coordinated student Participation** across **500+ residents**, fostering teamwork and engagement in hall activities
 - Organized and supported technical events and initiatives**, encouraging students to participate and collaborate in technology -driven activities
 - Coordinated technical resources, student teams**, and event logistics to ensure effective execution of technology-driven activities
- Teaching Assistant | Aquaculture Laboratory** (sep' 26-Present)
- Conducted laboratory sessions and guided first-year PG student in **aquaculture experiments, and practical problem-solving**
 - Supported hands-on training in **SolidWork, ANSYS, Python, and QGIS** for engineering analysis and visualization
 - Assisted students in experimental setup, data collection**, analysis, and troubleshooting during laboratory work

AWARDS AND ACHIEVEMENTS

National AI/ML Hackathon 2026-Actively participated in the National AI/ML Hackathon, Jointly organized by the Ministry of panchayati Raj, NIC Geospatial Division, and IIT Triupati

EXTRA CURRICULAR ACTIVITIES

- Volunteered | Smart India Hackathon (SIH) 2025-Hardware Edition**, supporting the organization and smooth execution of the hackathon coordinated technical and event- related activities (Dec'25)
- Volunteering with NSS** for a 7-days educational initiative, teaching **visually impaired students** and supporting their learning and development
- Inter-College Cricket Tournament winner**, Represented the college team and secured 1st position in an inter- college level cricket tournament
- Represent ABV at general championship IIT -Kharagpur in basketball, cricket, and football